

Chapter 7 Probability Distributions Assessment

“A reasonable probability is the only certainty.”

E.W. Howe

7.1	Introduction.....	2
7.2	Assessing Probabilities from Experts	3
7.2.1	Eliciting Discrete Probability Distribution from an Expert	3
7.2.2	Eliciting Continuous Probability Distribution from an Expert	4
7.3	Learning Probability Distributions from Data.....	5
7.3.1	Relative Frequencies Method.....	5
7.3.2	Some Common Probability Distributions	6
7.3.3	Method for Fitting Distribution to Data	12
7.3.4	Fitting Normal Distribution to Percentile Data.....	15
7.4	Discretization of Continuous Uncertain Variables	18
7.4.1	Discrete Approximation of PDF	18
7.4.2	Discrete Approximation of Some Common Distributions.....	20
7.4.3	Three-Point Quick Approximation Methods	25
	References.....	26
	Exercises	27

7.1 Introduction

- To perform Decision Analysis, we require the probability distributions of all the uncertain variables in the decision model.
- These probability distributions may be obtained by the following methods:
 1. Assessed directly via **Experts' Judgments** using **Probability Assessment Protocols**.
 2. **Learned from Data** using *Statistical or Machine Learning*.
 3. **Generated** from *Stochastic Models* such as queuing theory, Monte Carlo simulation, discrete event simulation, Markov chain, etc. But these models also require probability distribution assessments.
 4. Combinations of any of the above.
- We will cover methods 1 and 2 briefly in this chapter.
- Method 3 is covered in other IE courses.
- Note that although we have adopted the subjective view of the meaning of probabilities, we may still combine probabilities obtained objectively from data/models with those elicited directly from experts using the rules of Probability Theory.

7.2 Assessing Probabilities from Experts

7.2.1 Eliciting Discrete Probability Distribution from an Expert

- Eliciting a discrete probability distribution from an expert is a structured process used when empirical data is limited or unavailable. The goal is to translate the subjective judgment of the expert into a probability model while minimizing cognitive bias and inconsistency.
- For a discrete variable, the method will produce a probability mass function (PMF) for the variable.
- Let X be a discrete uncertain variable with two outcomes: x_1 and x_2 .
- The steps to assess the probability distribution of X are as follows:

1. Let W and L be two rewards such that $W \succ L$.

For example, W could be a small but significant cash amount (e.g., \$100), and L is nothing.

2. Set the probability wheel with the orange sector at p .

3. Repeat

4. Ask the expert to choose between the two options:

A: Spin the probability wheel.

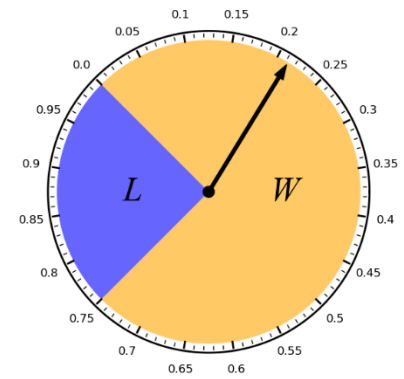
If the outcome is orange, he receives W .

Otherwise, he receives L .

B: Do not spin the wheel.

If the outcome of $X = x_1$, he receives W .

Otherwise, he receives L .



5. If the expert chose *A*, reduce the value of p set on the wheel.
If the expert chose *B*, increase the value of p set on the wheel.

6. **Until** the expert is indifferent between options *A* and *B*.

7. Result: $\Pr \{X = x_1\} = p$
 $\Pr \{X = x_2\} = 1 - p.$

Discrete variables with more than 2 outcomes:

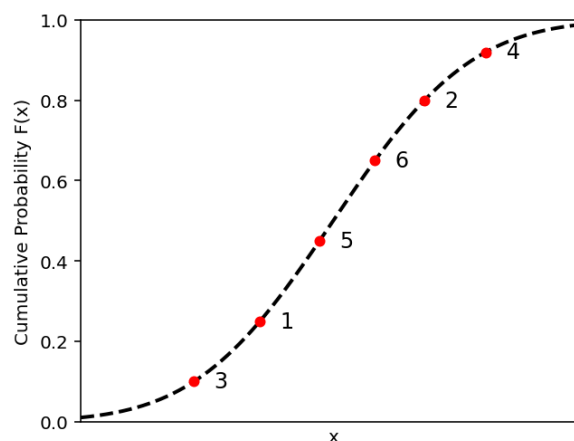
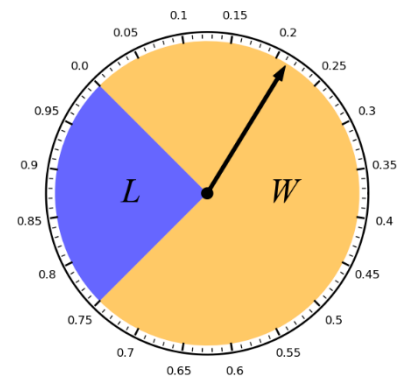
- If X has more than two discrete outcomes, the procedure above is repeated for outcomes x_2 to x_{n-1} to obtain $p_2, \dots, p_{n-1}$, and let $p_n = 1 - \sum_{i=1}^{n-1} p_i$.
- The output is a probability mass function for X .

7.2.2 Eliciting Continuous Probability Distribution from an Expert

- For a continuous uncertain variable, the method will produce a cumulative distribution function (CDF) for the variable.
- Let X be a continuous uncertain variable. The steps to assess its CDF are as follows:
 - Let W and L be two rewards such that $W \succ L$.
For example, W could be a small but significant cash amount (e.g., \$100) and L is nothing.
 - Select a value of x from the range of possible values of X based on a selected assessment protocol.
 - Set the probability wheel with the orange sector at p .
 - Repeat**
 - Ask the expert to choose between the two options:

A: Spin the probability wheel.
If the outcome is orange, he receives W .
Otherwise, he receives L .

B: Do not spin the wheel.
If the actual outcome of $X \leq x$, he receives W .
Otherwise, he receives L .
 - If the expert chose *A*, reduce the value of p set on the wheel.
If the expert chose *B*: increase the value of p set on the wheel.
 - Until** the expert is indifferent between options *A* and *B*.
 - Encode $\Pr(X \leq x) = p$ on the CDF for X .
 - Repeat Steps 1 to 8 for a number of other selected values of x based on the selected assessment protocols. Each of these values results in a point on the CDF for X .
 - Plot the CDF for X .



- Note: The choice and sequence of x values are important to avoid biases in the results. See Stanford/SRI protocol for guidelines on how to choose the sequence of values.

7.3 Learning Probability Distributions from Data

- We may make use of data to access a probability distribution.
- Some basic methods of using data for assessing probability distributions are discussed in this section.

7.3.1 Relative Frequencies Method

- The **relative frequency** of past events is often a good starting point to obtain an estimate of the probabilities of a discrete variable with few outcomes.

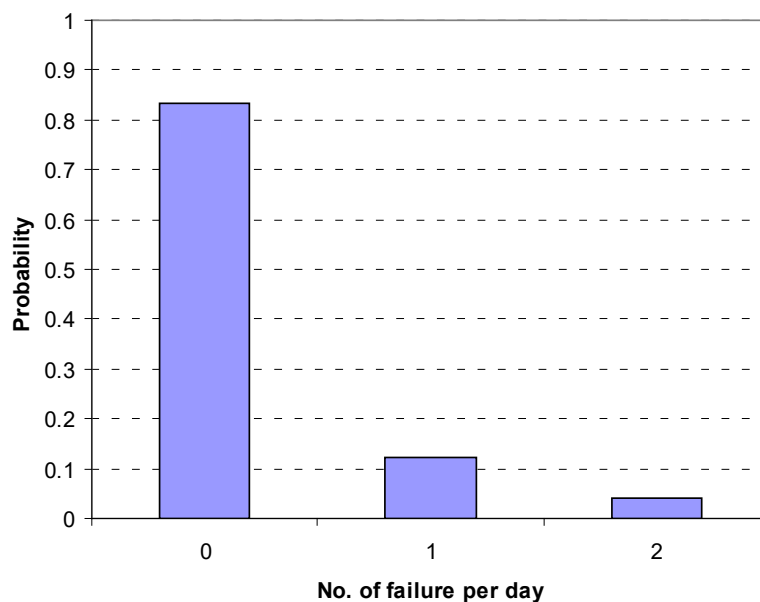
Example

- Let X = the number of machine failures per day.
- The following data were collected over a period of 260 days:

No. of failures	No. of occurrences
0	217
1	32
2	11

- The probability mass function for X may be approximated by:

No. of Failures	Probability
0	$217/260 = 0.835$
1	$32/260 = 0.123$
2	$11/260 = 0.042$



7.3.2 Some Common Probability Distributions

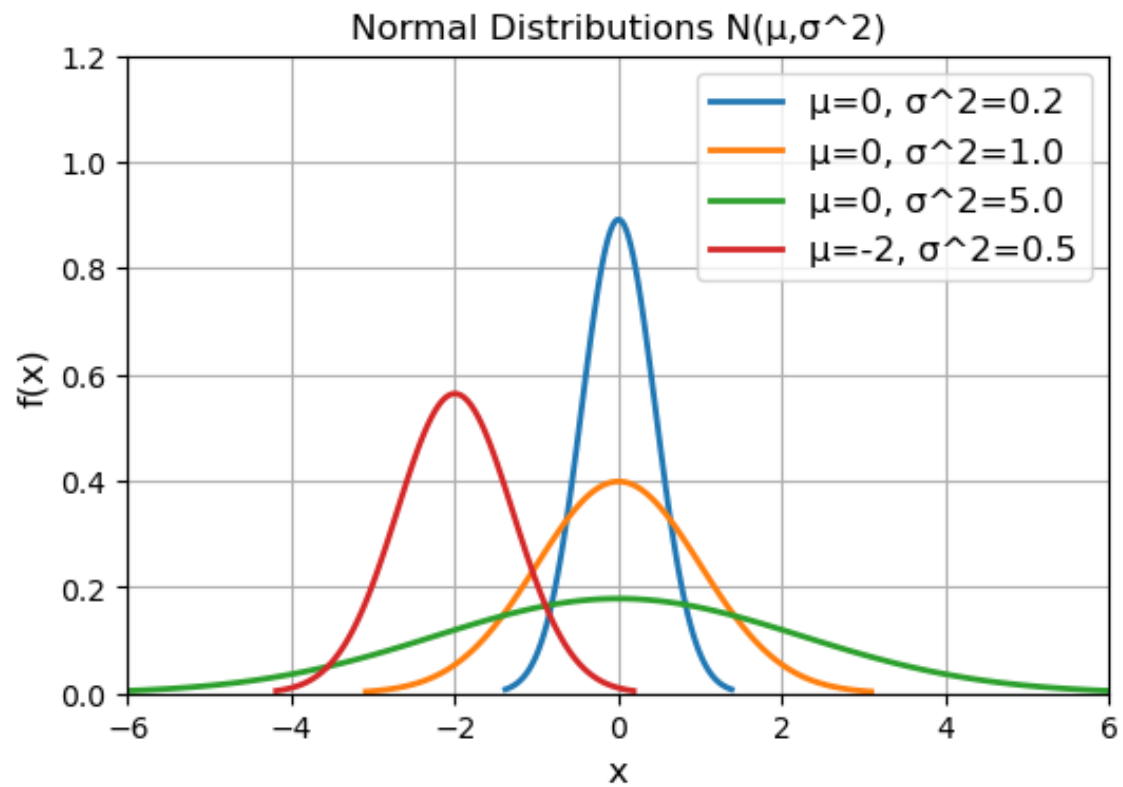
Normal Distribution: $N(\mu, \sigma^2)$

- The most commonly used distribution.

$$\text{PDF: } f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-1/2((x-\mu)^2 / \sigma^2)} \text{ for } -\infty < x < \infty$$

μ = mean, σ = standard deviation,

location = μ , *scale* = σ

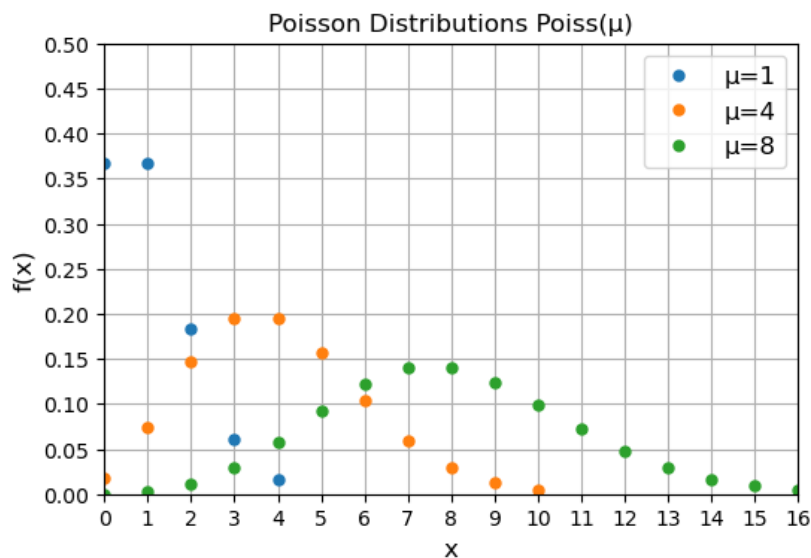


Poisson Distribution: $\text{Pois}(\mu)$

- The Poisson distribution is a **discrete distribution** for modeling the number of occurrences in a given time interval.

$$\text{PMF: } f(x; \mu) = \begin{cases} \frac{\mu^x e^{-\mu}}{x!} & \text{if } x = 1, 2, 3, \dots \\ 0 & \text{Otherwise} \end{cases}$$

μ = average per unit time, variance = μ , location = 0, scale = μ .

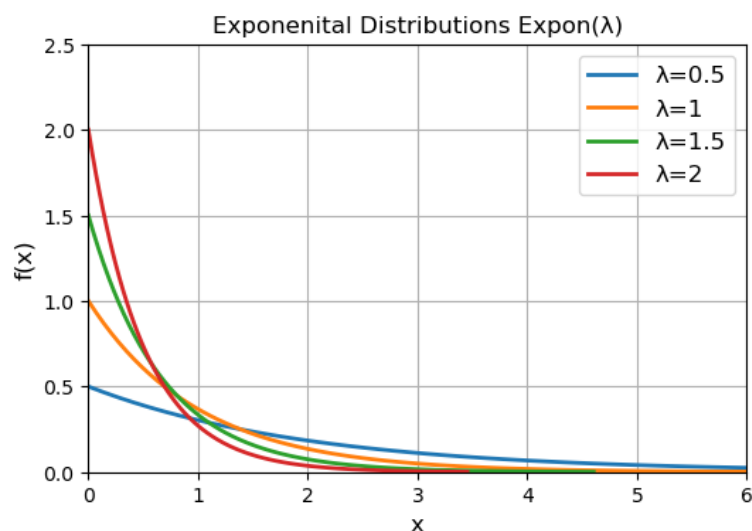


Exponential Distribution: $\text{Expon}(\lambda)$

- The exponential distribution is popular for modeling the inter-arrival time of a Poisson process.

$$\text{PDF: } f(x; \lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

where $\lambda > 0$ is the rate parameter, mean = $1/\lambda$, variance = $1/\lambda^2$, location = 0, scale = $1/\lambda$

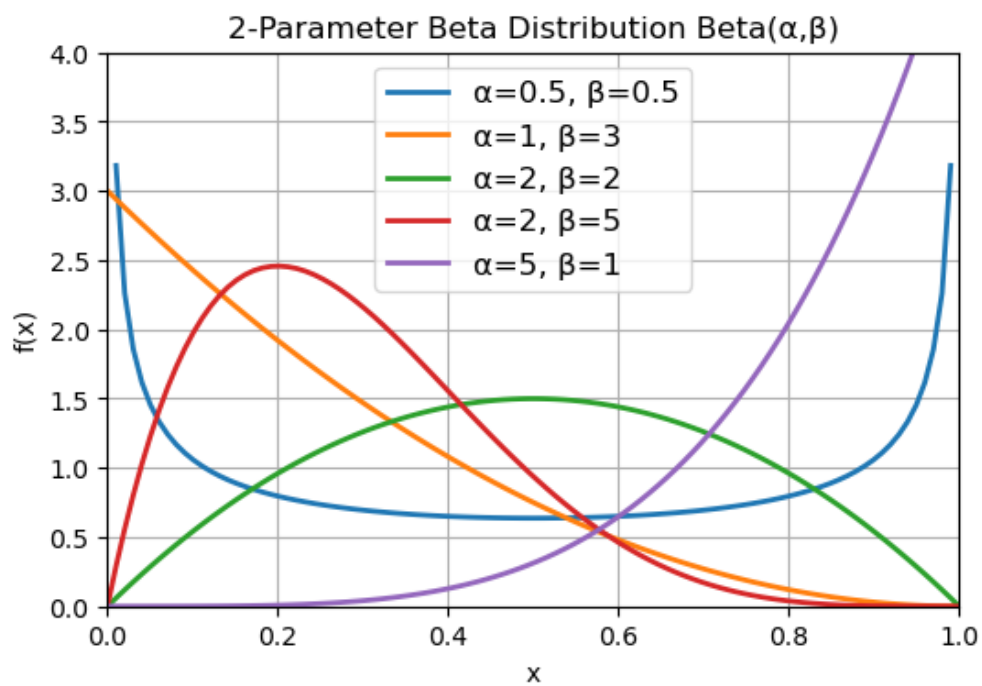


Standard (2-parameter) Beta Distribution: $\text{Beta}(\alpha, \beta)$.

- The standard 2-parameter Beta distribution is often used for modeling the uncertainty over the probability p of an event where $0 \leq p \leq 1$.

$$\text{PDF: } f(x; \alpha, \beta) = \begin{cases} \frac{x^{\alpha-1}(1-x)^{\beta-1}}{\int_0^1 t^{\alpha-1}(1-t)^{\beta-1} dt} & 0 \leq x \leq 1 \\ 0 & \text{Otherwise} \end{cases}$$

$\alpha > 0$ and $\beta > 0$ are shape parameters, mean = $\frac{\alpha}{\alpha + \beta}$, variance = $\frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$



General (4-parameter) Beta Distribution: Beta (α, β, L, H)

- The standard (2-parameter) Beta distribution is defined over a variable x where $0 \leq x \leq 1$.
- We can use the general (4-parameter) Beta distribution when $L \leq x \leq H$, by applying the linear transformation $y = x(H - L) + L$.
- Y follows Beta (α, β, L, H) if and only if X follows Beta (α, β) where $X = (Y - L)/(H - L)$.

$$\text{PDF: } f(y; \alpha, \beta, L, H) = \begin{cases} \frac{f(x; \alpha, \beta)}{(H - L)} & L \leq y \leq H \\ 0 & \text{Otherwise} \end{cases}$$

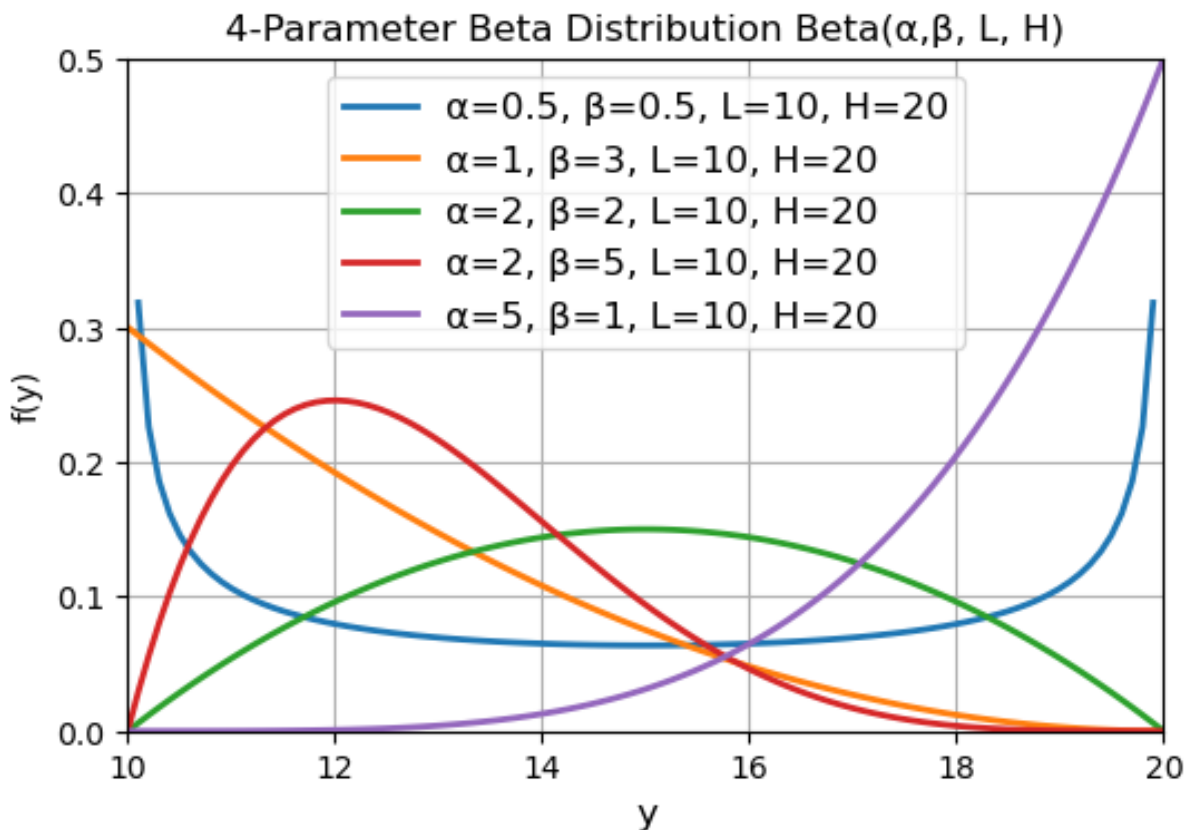
where $x = \frac{(y - L)}{(H - L)}$ and $\alpha > 0$ and $\beta > 0$ are shape parameters.

$$\text{mean} = \frac{\alpha H + \beta L}{\alpha + \beta}$$

$$\text{variance} = \frac{\alpha \beta (H - L)^2}{(\alpha + \beta)^2 (\alpha + \beta + 1)}$$

$$\text{location} = L$$

$$\text{scale} = (H - L)$$



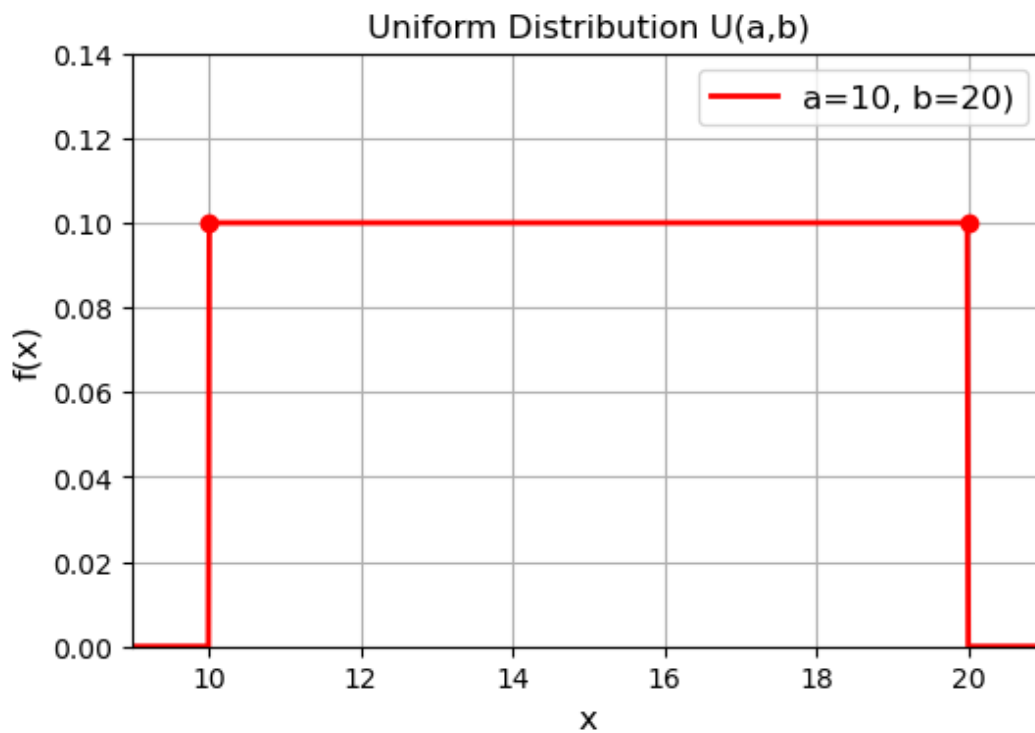
General Uniform Distribution: $U(a, b)$

- We can use the uniform distribution when we are totally uncertain about an event and have completely no other information about it.
- The uniform distribution represents the state of highest uncertainty over an event.

$$\text{PDF: } f(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

$$\text{mean} = \frac{(a+b)}{2}, \text{variance} = \frac{(b-a)^2}{12}$$

$$\text{Location} = a, \text{Scale} = (b - a)$$



Triangular Distribution: Triang(a, b, m)

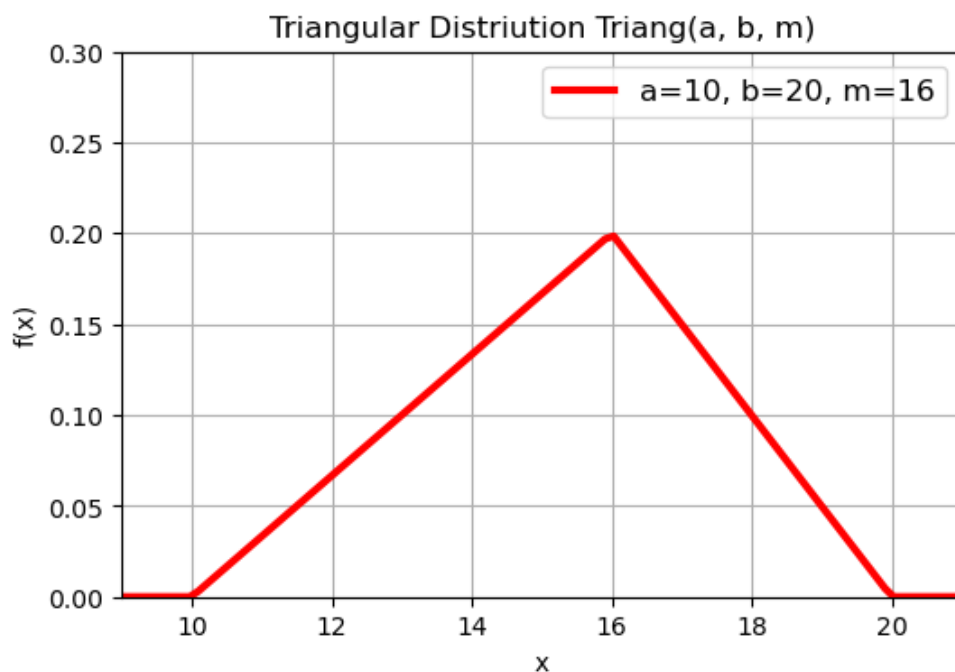
- The triangular distribution may be used to approximate a variable that is known to be bounded between a and b , is asymmetrical, and has the highest chance of occurrences around the mode m .

$$\text{PDF: } f(x) = \begin{cases} 0 & x < a \\ \frac{2(x-a)}{(b-a)(m-a)} & a \leq x < m \\ \frac{2}{b-a} & x = m \\ \frac{2(b-x)}{(b-a)(b-m)} & m < x \leq b \\ 0 & b < x \end{cases}$$

a = lower bound, b = upper bound, m = mode.

$$\text{mean} = \frac{(a+b+m)}{3}, \quad \text{variance} = \frac{a^2 + b^2 + m^2 - ab - am - bm}{18},$$

$$\text{Location} = a, \quad \text{Scale} = \frac{m-a}{b-a}$$



7.3.3 Method for Fitting Distribution to Data

The Maximum Likelihood Estimation (MLE) Method

- This is a popular parameter estimation method and is easy to implement computationally.
- Let $\mathbf{D} = \{x_1, x_2, \dots, x_n\}$ be a set of observed data values. Suppose we want to fit data $\mathbf{D}$ to a probability distribution with PDF or PMF $f(x|\theta)$ where θ is a vector of unknown parameters.
- The **Likelihood Function** $L(\theta | \mathbf{D})$ is the joint probability of observing all the x values in $\mathbf{D}$ given that the distribution to be fitted has the parameter θ . That is,

$$L(\theta | \mathbf{D}) = f(x_1, x_2, \dots, x_n | \theta)$$

- The MLE method finds the value of θ that maximizes $L(\theta | \mathbf{D})$, that is

$$\hat{\theta} = \arg \text{Max}_{\theta} f(x_1, x_2, \dots, x_n | \theta)$$

- If the data values are assumed to be mutually independent, then

$$\hat{\theta} = \arg \text{Max}_{\theta} \left(\prod_{i=1}^n f(x_i | \theta) \right)$$

- It is usually more convenient to **Minimize the Negative Log-Likelihood Function** instead, as $\log()$ is a monotonically increasing function. That is

$$\hat{\theta} = \arg \text{Min}_{\theta} \left(- \sum_{i=1}^n \log f(x_i | \theta) \right)$$

- Some distributions (e.g., Normal, Uniform, Exponential, Poisson, Gamma) have known closed-form optimal solutions for $\hat{\theta}$. Otherwise, numerical optimization algorithms are often used to find $\hat{\theta}$.

Using Software Tools

1. Excel add-ins.

- Real Statistics Resource Pack (Free) (<https://real-statistics.com/>)
- Simulación 5.0 (Free) (<https://sites.google.com/view/simulacion5/main>)
- XLSTAT (Free Trial) (<https://www.xlstat.com/>)
- EasyFit (Free Trial) (<https://easyfit.en.softonic.com/>)
- RiskAMP (Free Trial) (<https://www.riskamp.com/>)
- @Risk (Free Trial) (<https://lumisivero.com/>)

2. Python Module: `scipy.stats`

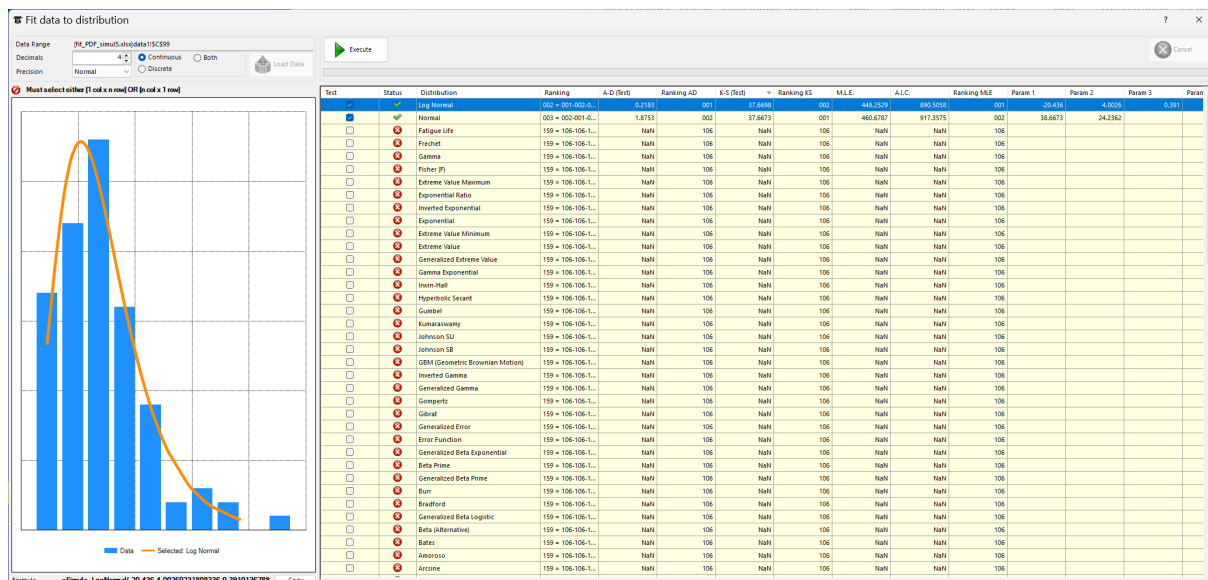
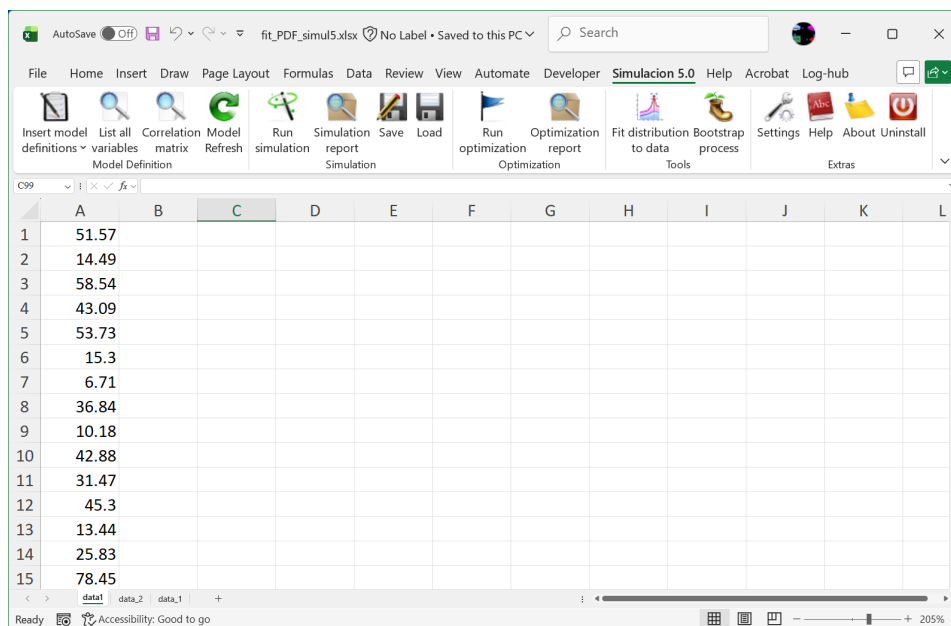
- **Continuous distributions:** Direct support using MLE (maximum likelihood estimation) and MM (method of moments).
- **Discrete distributions:** No direct support. Write your own code to minimize the negative log-likelihood function to fit the parameters.

Example 1 (Fitting Continuous Distributions)

- Fit a continuous distribution to the following data set (100 observations):

51.57	31.47	30.80	34.93	92.05	54.44	19.21	46.26	39.68	5.63
14.49	45.3	87.35	33.42	20.14	46.72	22.51	23.06	28.88	14.65
58.54	13.44	11.27	38.4	53.02	32.14	28.72	29.43	23.62	63.97
43.09	25.83	15.34	52.34	67.89	59.24	17.39	28.12	26.59	32.81
53.73	78.45	45.55	21.27	34.39	107.04	37.82	30.55	31.64	45.93
15.30	24.90	60.03	41.65	34.04	65.43	27.18	18.4	44.21	34.35
6.71	27.79	29.59	40.76	37.69	34.06	1.48	33.48	35.09	143.46
36.84	44.17	64.63	57.48	81.72	11.93	10.70	20.30	9.12	38.46
10.18	90.22	15.10	23.18	65.84	12.14	1.83	41.85	48.65	61.83
42.88	39.75	23.54	36.98	43.19	21.06	104.88	46.90	1.53	54.13

Using Simulacion 5.0 -> Fit distribution to data



LogNormal(-20.436, 4.00260231808336, 0.391013678821476)
Normal(38.6673, 24.2362)

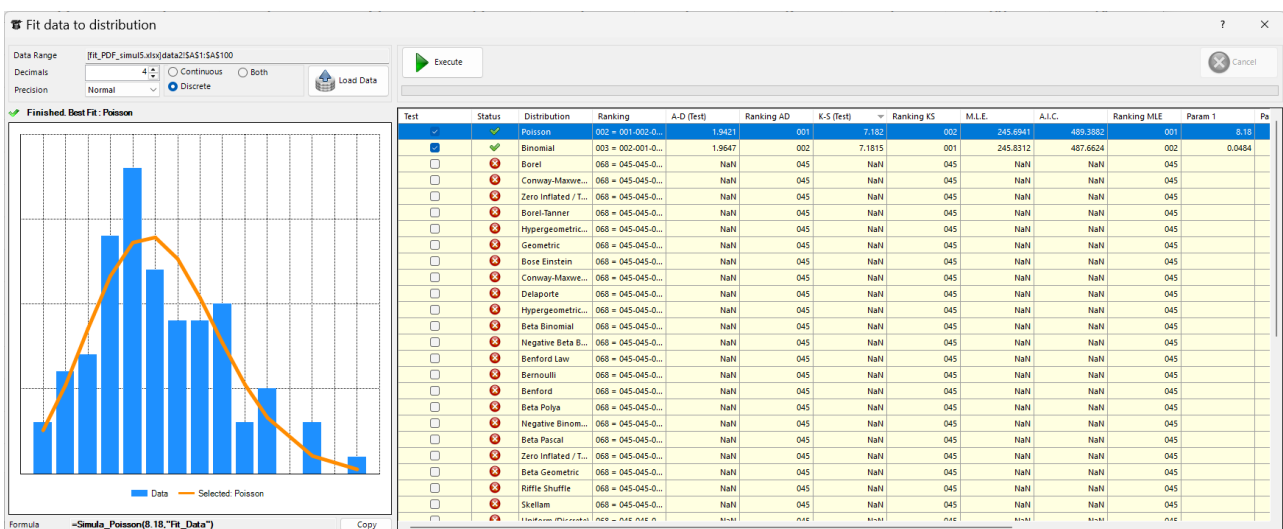
Example 2 (Fitting Discrete Distributions)

- Fit a distribution to the following 100 values observed from a discrete counting event.

12	11	10	15	3	11	7	8	8	8
7	4	11	8	4	7	6	3	7	6
4	7	6	9	6	8	6	7	9	8
6	13	10	7	5	10	7	11	12	8
9	8	5	9	7	7	7	3	7	7
9	10	10	10	13	5	11	8	11	6
4	6	11	5	5	8	13	11	7	6
13	9	6	15	10	13	7	7	17	5
7	7	9	15	8	4	10	5	8	10
6	9	12	6	4	6	6	11	9	11

Using Simulacion 5.0

Fitting a Poisson and a Binomial distribution

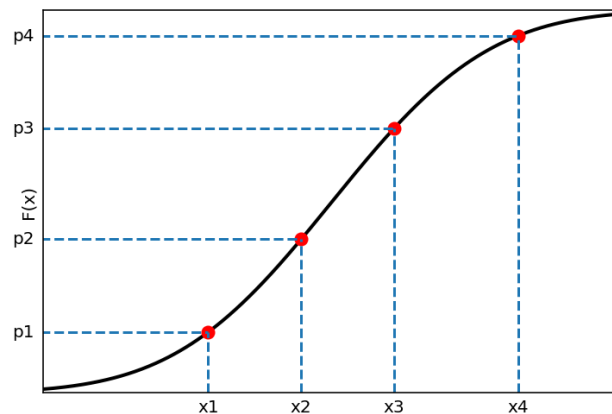


Poisson(8.18)

Binomial(0.0484,169)

7.3.4 Fitting Normal Distribution to Percentile Data

- Suppose an uncertain variable X follows the Normal distribution with several known percentile values as shown in the following CDF:



- Given the data $x_1, x_2, \dots, x_n$ and their corresponding percentile values $p_1, p_2, \dots, p_n$:

$$F(x_i | \mu, \sigma) = p_i \quad \text{for } i = 1, 2, \dots, n$$

$$\Phi((x_i - \mu)/\sigma) = p_i \quad \text{for } i = 1, 2, \dots, n$$

where Φ is the CDF of the standard normal distribution $N(0, 1)$

- We may find μ and σ by solving the above n non-linear equations. However, the inverse forms of these equations are linear and easy to solve.

$$x_i = \mu + \Phi^{-1}(p_i) \sigma \quad \text{for } i = 1, 2, \dots, n$$

$$x_i = \mu + z_i \sigma \quad \text{for } i = 1, 2, \dots, n$$

where $z_i = \Phi^{-1}(p_i)$ is the quantile function or the percent-point function (PPF) of the standard normal distribution.

- The value of z_i can be computed:
 - Python: `z_i = scipy.stats.norm.ppf(p_i)`
 - Excel: `=NORM.S.INV(p_i)` or `=NORM.INV(p_i, 0, 1)`
- When $n=2$, we have two linear equations with closed-form solutions for μ and σ .

$$x_1 = \mu + z_1 \sigma$$

$$x_2 = \mu + z_2 \sigma$$

$$\sigma = \frac{x_2 - x_1}{z_2 - z_1} \quad \text{and} \quad \mu = x_1 - z_1 \sigma$$

- When $n > 2$, we can use (least-squares errors) linear regression to find the best fit for the parameters μ and σ .

Example

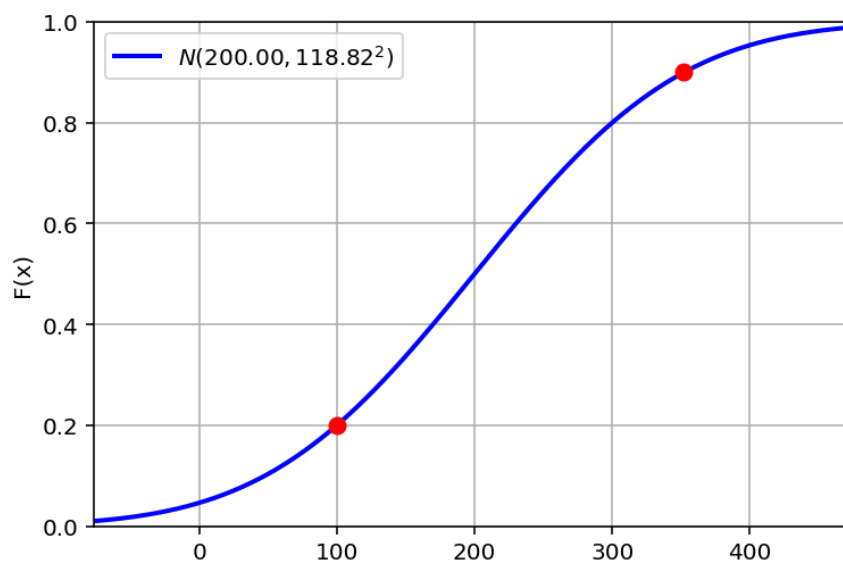
- An uncertain variable X follows the normal distribution. It is observed that 10% of its values are greater than or equal to 352.27 and 20% of its values are less than or equal to 100.0. What are the mean and standard deviation of X ?
- Given
$$\begin{array}{ll} x_1 = 352.27 & p_1 = 1 - 0.1 = 0.9 \\ x_2 = 100 & p_2 = 0.2 \end{array}$$

$$z_1 = \Phi^{-1}(0.9) = 1.281552$$

$$z_2 = \Phi^{-1}(0.2) = -0.84162$$

$$\sigma = (352.27 - 100) / (1.28155 - (-0.84162)) = 118.817$$

$$\mu = 352.27 - 1.281552 \times 118.817 = 200.0$$



- Python code:

```
# Fit normal distribution to 2 percentile values
```

```
import numpy as np
from scipy.stats import norm
import matplotlib.pyplot as plt
```

```
# Percentile data
```

```
x = np.array([100, 352.27])
p = np.array([0.2, 0.9])
```

```
# Compute mu and sigma
```

```
z = norm.ppf(p)
sigma = (x[1]-x[0])/(z[1]-z[0])
mu = x[0] - z[0]*sigma
print(f"mu = {mu:.2f}")
print(f"Sigma = {sigma:.2f}")
```

```
mu = 200.00
```

```
Sigma = 118.82
```



```

# Plot the CDF of the fitted normal distribution
xmin = norm.ppf(0.01, loc=mu, scale=sigma)
xmax = norm.ppf(0.99, loc=mu, scale=sigma)
xp = np.linspace(xmin, xmax, 500)
yp = norm.cdf(xp, mu, sigma)

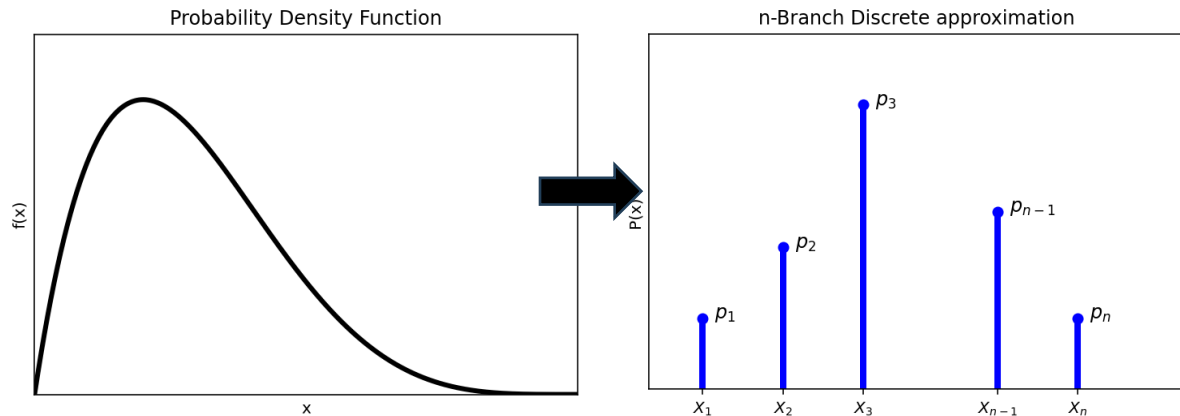
fig, ax = plt.subplots()
ax.plot(xp, yp, lw=2, color='blue', label=fr"$N(\{\mu:.2f\}, \{\sigma:.2f\}^2)$")
ax.plot(x, p, 'ro', ms=7)
ax.set_xlim(xmin, xmax)
ax.set_ylim(0, 1)
ax.set_ylabel("F(x)")
ax.legend()
ax.grid()
plt.show()

```

7.4 Discretization of Continuous Uncertain Variables

7.4.1 Discrete Approximation of PDF

- Suppose we have a continuous uncertain variable with a known PDF, and we wish to represent it in a decision tree or influence diagram with a discrete chance node.
- We need to find the *best approximate discrete probability distribution* with n outcomes, i.e., convert the PDF $f(x)$ into an ‘equivalent’ PMF = $\{(x_i, p_i)\}_{i=1}^n$. There are $2n$ unknowns to be determined.



Desired Properties

- Given a continuous variable X with PDF $f(x)$, we seek to approximate it by a probability mass function (PMF)

$$\hat{X} = \{(x_i, p_i)\}_{i=1}^n \text{ such that } \sum_{i=1}^n p_i = 1.$$

- A desired property is that the expected value of *any function* $g(x)$ should be preserved across the discretization. That is

$$\int_{-\infty}^{+\infty} g(x) f(x) dx = \sum_{i=1}^n p_i g(x_i) \text{ for all functions } g(x).$$

- If $g(x)$ is the utility function, then the expected utility will be preserved, and there will be no error in the optimal decision policy computation due to the discretization.
- Assume that $g(x)$ can be approximated by the polynomial $g(x) = a_0 + a_1x + a_2x^2 + \dots$. We therefore require that

$$\begin{aligned} \int_{-\infty}^{\infty} (a_0 + a_1x + a_2x^2 + \dots) f(x) dx &= \sum_{i=1}^n p_i (a_0 + a_1x_i + a_2x_i^2 + \dots) \\ a_0 + a_1E[x] + a_2E[x^2] + \dots &= a_0 + a_1 \sum_{i=1}^n p_i x_i + a_2 \sum_{i=1}^n p_i x_i^2 + \dots \end{aligned}$$

- Equating term by term:

$$E[x^k] = \sum_{i=1}^n p_i x_i^k \quad \text{for } k = 0, 1, 2, \dots$$

- Hence, all moments must be preserved between the continuous and the discrete distributions:

$$k=0 \quad \Rightarrow \quad \sum_{i=1}^n p_i = 1 \quad \text{discrete distribution must be valid.}$$

$$k=1 \quad \Rightarrow \quad E[x] = \sum_{i=1}^n p_i x_i \quad \text{mean must be preserved.}$$

$$k=2 \quad \Rightarrow \quad E[x^2] = \sum_{i=1}^n p_i x_i^2 \quad \text{variance must be preserved}$$

(if the mean is already preserved)

$$k=3 \quad \Rightarrow \quad E[x^3] = \sum_{i=1}^n p_i x_i^3 \quad \text{skewness must be preserved.}$$

(if the mean is already preserved)

Limit the number of moments that can be preserved

- In principle, the discrete distribution must preserve all the moments, i.e.,

$$E[x^k] = \sum_{i=1}^n p_i x_i^k = \int x^k f(x) dx \quad \text{for all } k = 0, 1, 2, \dots$$

- However, we can only solve the first $2n$ equations to find the $2n$ unknowns x_i, p_i for $i = 1$ to n .
- Hence, only the first $(2n - 1)$ moments can be preserved by solving:

$$\sum_{i=1}^n p_i = 1 \quad (1)$$

$$\int x f(x) dx = \sum_{i=1}^n p_i x_i \quad (2)$$

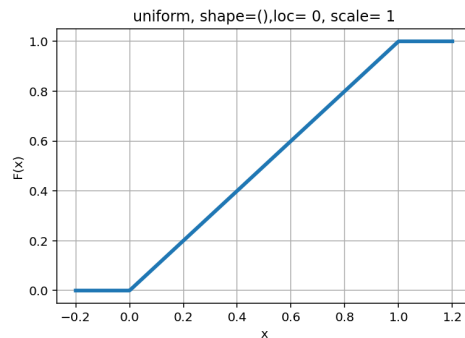
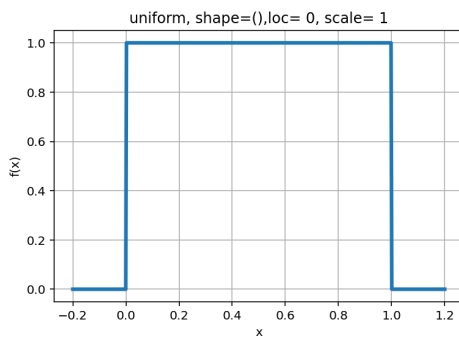
$$\int x^2 f(x) dx = \sum_{i=1}^n p_i x_i^2 \quad (3)$$

$$\dots$$

$$\int x^{2n-1} f(x) dx = \sum_{i=1}^n p_i x_i^{2n-1} \quad (2n)$$

7.4.2 Discrete Approximation of Some Common Distributions

Standard Uniform Distribution $u(0, 1)$

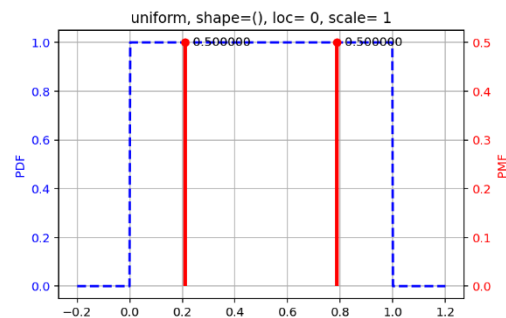


- To approximate $u(0,1)$ with a 2-branch probability distribution, we solve the following 4 equations to match the first 3 moments:

$$\begin{aligned} p_1 + p_2 &= 1 \\ x_1 p_1 + x_2 p_2 &= \int_0^1 x dx = \frac{1}{2} \\ x_1^2 p_1 + x_2^2 p_2 &= \int_0^1 x^2 dx = \frac{1}{3} \\ x_1^3 p_1 + x_2^3 p_2 &= \int_0^1 x^3 dx = \frac{1}{4} \end{aligned}$$



$x_1 = 0.211325$	$p_1 = 0.5$
$x_2 = 0.788675$	$p_2 = 0.5$

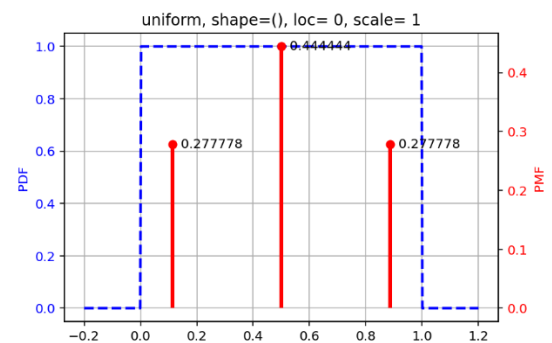


- To approximate $u(0,1)$ with a 3-branch probability distribution, we solve the following 6 equations to match the first 5 moments:

$$\begin{aligned} p_1 + p_2 + p_3 &= 1 \\ x_1 p_1 + x_2 p_2 + x_3 p_3 &= \int_0^1 x dx = \frac{1}{2} \\ x_1^2 p_1 + x_2^2 p_2 + x_3^2 p_3 &= \int_0^1 x^2 dx = \frac{1}{3} \\ x_1^3 p_1 + x_2^3 p_2 + x_3^3 p_3 &= \int_0^1 x^3 dx = \frac{1}{4} \\ x_1^4 p_1 + x_2^4 p_2 + x_3^4 p_3 &= \int_0^1 x^4 dx = \frac{1}{5} \\ x_1^5 p_1 + x_2^5 p_2 + x_3^5 p_3 &= \int_0^1 x^5 dx = \frac{1}{6} \end{aligned}$$



$x_1 = 0.112702$	$p_1 = 0.277778$
$x_2 = 0.5$	$p_2 = 0.444444$
$x_3 = 0.887298$	$p_3 = 0.277778$

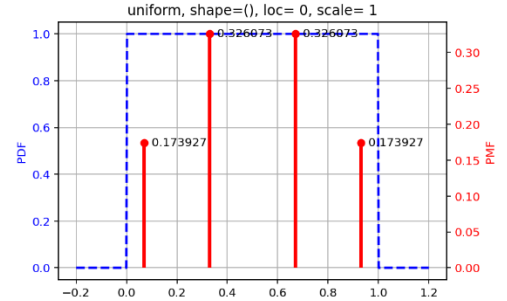


- To approximate $u(0,1)$ with a 4-branch probability distribution, we solve the following 8 equations to match the first 7 moments:

$$\begin{aligned}
 p_1 + p_2 + p_3 + p_4 &= 1 \\
 x_1 p_1 + x_2 p_2 + x_3 p_3 + x_4 p_4 &= \int_0^1 x dx = \frac{1}{2} \\
 &\vdots \\
 x_1^6 p_1 + x_2^6 p_2 + x_3^6 p_3 + x_4^6 p_4 &= \int_0^1 x^6 dx = \frac{1}{7} \\
 x_1^7 p_1 + x_2^7 p_2 + x_3^7 p_3 + x_4^7 p_4 &= \int_0^1 x^7 dx = \frac{1}{8}
 \end{aligned}$$



$x_1 = 0.069432$	$p_1 = 0.173927$
$x_2 = 0.330009$	$p_2 = 0.326073$
$x_3 = 0.669991$	$p_3 = 0.326073$
$x_4 = 0.930568$	$p_4 = 0.173927$

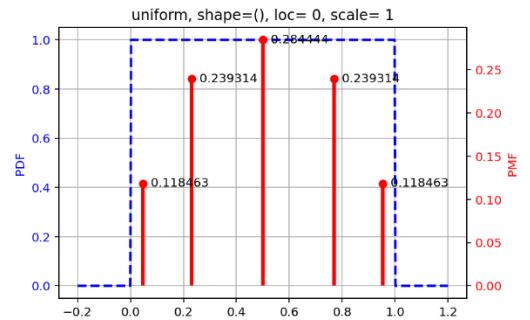


- To approximate $u(0,1)$ with a 5-branch probability distribution, we solve the following 10 equations to match the first 9 moments:

$$\begin{aligned}
 p_1 + p_2 + p_3 + p_4 + p_5 &= 1 \\
 x_1 p_1 + x_2 p_2 + x_3 p_3 + x_4 p_4 + x_5 p_5 &= \int_0^1 x dx = \frac{1}{2} \\
 &\vdots \\
 x_1^8 p_1 + x_2^8 p_2 + x_3^8 p_3 + x_4^8 p_4 + x_5^8 p_5 &= \int_0^1 x^8 dx = \frac{1}{9} \\
 x_1^9 p_1 + x_2^9 p_2 + x_3^9 p_3 + x_4^9 p_4 + x_5^9 p_5 &= \int_0^1 x^9 dx = \frac{1}{10}
 \end{aligned}$$



$x_1 = 0.046910$	$p_1 = 0.118463$
$x_2 = 0.230765$	$p_2 = 0.239314$
$x_3 = 0.5$	$p_3 = 0.284444$
$x_4 = 0.769235$	$p_4 = 0.239314$
$x_5 = 0.95309$	$p_5 = 0.118463$



General Uniform Distribution $u(a, b)$

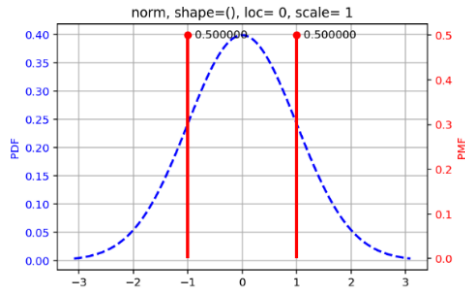
- location = a , scale = $(b - a)$
- The n -branch discrete approximation for $u(a, b)$ is $\{(y_i, p_i)\}_{i=1}^n$

where $y_i = a + (b - a) x_i$

and $\{(x_i, p_i)\}_{i=1}^n$ is the n -branch discrete approximation for $u(0, 1)$.

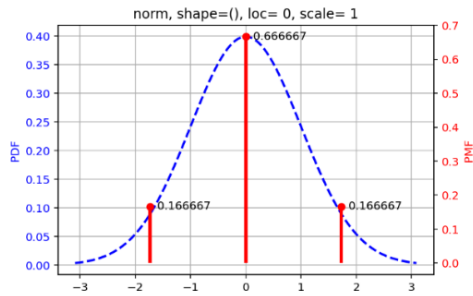
Standard Normal Distribution $N(0, 1)$

- 2-branch probability distribution preserving the first 3 moments:



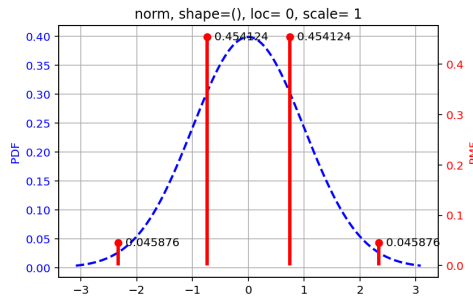
$x_1 = -1$	$p_1 = 0.5$
$x_2 = +1$	$p_2 = 0.5$

- 3-branch probability distribution preserving the first 5 moments:



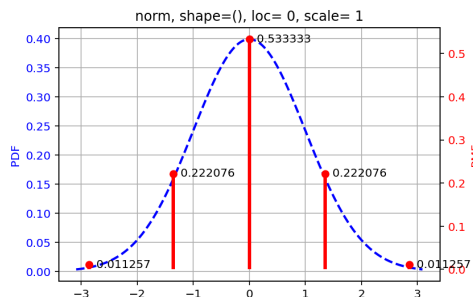
$x_1 = -1.732051$	$p_1 = 1/6$
$x_2 = 0$	$p_2 = 2/3$
$x_3 = 1.732051$	$p_3 = 1/6$

- 4-branch probability distribution preserving the first 7 moments:



$x_1 = -2.334414$	$p_1 = 0.045876$
$x_2 = -0.741964$	$p_2 = 0.454124$
$x_3 = 0.741964$	$p_3 = 0.454124$
$x_4 = 2.334414$	$p_4 = 0.045876$

- 5-branch probability distribution preserving the first 9 moments:



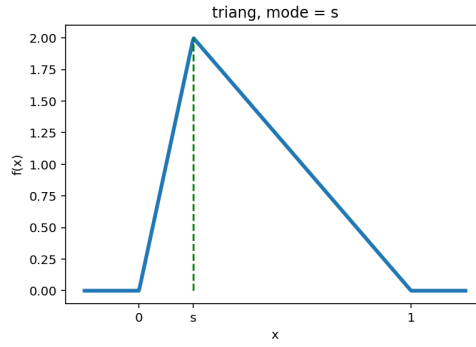
$x_1 = -2.856970$	$p_1 = 0.011257$
$x_2 = -1.355626$	$p_2 = 0.222076$
$x_3 = 0$	$p_3 = 0.533333$
$x_4 = 1.355626$	$p_4 = 0.222076$
$x_5 = 2.856970$	$p_5 = 0.011257$

General Normal Distribution $N(\mu, \sigma^2)$

- location = μ , scale = σ .
- The n -branch discrete approximation for $N(\mu, \sigma^2)$ is $\{(y_i, p_i)\}_{i=1}^n$
 where $y_i = \mu + \sigma z_i$ for $i = 1, \dots, n$
 and $\{(z_i, p_i)\}_{i=1}^n$ is the n -branch discrete approximation for $N(0, 1)$

Triangular Distribution Triang(0, 1, s)

- The Triang(0, 1, s) distribution has support $[0, 1]$ and mode $0 \leq s \leq 1$.



- The discrete n -branch approximations for Triang(0, 1, s) for selected values of $n = 2$ to 5, and $s \in (0, 0.2, 0.25, 0.3, 1/3, 0.4, 0.5)$ are given in the tables below:

Triang(0, 1, 0)			
x1	0.155051	p1	0.636083
x2	0.644949	p2	0.363917

Triang(0, 1, 0.1)			
x1	0.194874	p1	0.631407
x2	0.660950	p2	0.368593

Triang(0, 1, 0.2)			
x1	0.229366	p1	0.615798
x2	0.673491	p2	0.384202

Triang(0, 1, 0.25)			
x1	0.244393	p1	0.603322
x2	0.678684	p2	0.396678

x1	0.088588	p1	0.401864
x2	0.409467	p2	0.458482
x3	0.787659	p3	0.139654

x1	0.127938	p1	0.387786
x2	0.432987	p2	0.469055
x3	0.796031	p3	0.143159

x1	0.154412	p1	0.346637
x2	0.447389	p2	0.500215
x3	0.801081	p3	0.153149

x1	0.163157	p1	0.319651
x2	0.453644	p2	0.520506
x3	0.803219	p3	0.159843

x1	0.057104	p1	0.271014
x2	0.276843	p2	0.406929
x3	0.583590	p3	0.259695
x4	0.860240	p4	0.062362

x1	0.093211	p1	0.245247
x2	0.301569	p2	0.420702
x3	0.597628	p3	0.269338
x4	0.864932	p4	0.064713

x1	0.109678	p1	0.188578
x2	0.316011	p2	0.450203
x3	0.605420	p3	0.291173
x4	0.867507	p4	0.070047

x1	0.113564	p1	0.163039
x2	0.324711	p2	0.459890
x3	0.609519	p3	0.303845
x4	0.868826	p4	0.073226

x1	0.039810	p1	0.193563
x2	0.198013	p2	0.334349
x3	0.437975	p3	0.292774
x4	0.695464	p4	0.147818
x5	0.901465	p5	0.031496

x1	0.071345	p1	0.156666
x2	0.221487	p2	0.348637
x3	0.454126	p3	0.306698
x4	0.704166	p4	0.154971
x5	0.904274	p5	0.033028

x1	0.080404	p1	0.103899
x2	0.238350	p2	0.360206
x3	0.464011	p3	0.331764
x4	0.709315	p4	0.168241
x5	0.905916	p5	0.035889

x1	0.082616	p1	0.088002
x2	0.248174	p2	0.351340
x3	0.468896	p3	0.346679
x4	0.711799	p4	0.176337
x5	0.906702	p5	0.037642

Triang(0, 1, 0.3)			
x1	0.257842	p1	0.587644
x2	0.683424	p2	0.412356

Triang(0, 1, 1/3)			
x1	0.265910	p1	0.575485
x2	0.686470	p2	0.424515

Triang(0, 1, 0.4)			
x1	0.279898	p1	0.547596
x2	0.692734	p2	0.452404

Triang(0, 1, 0.5)			
x1	0.295876	p1	0.500000
x2	0.704124	p2	0.500000

x1	0.169526	p1	0.291921
x2	0.460604	p2	0.540763
x3	0.805468	p3	0.167316

x1	0.172779	p1	0.274265
x2	0.465917	p2	0.553022
x3	0.807087	p3	0.172714

x1	0.177792	p1	0.243086
x2	0.478314	p2	0.572118
x3	0.810599	p3	0.184797

x1	0.183772	p1	0.208333
x2	0.500000	p2	0.583333
x3	0.816228	p3	0.208333

x1	0.116338	p1	0.142994
x2	0.334618	p2	0.462135
x3	0.613761	p3	0.318029
x4	0.870170	p4	0.076842

x1	0.117936	p1	0.132423
x2	0.341447	p2	0.459200
x3	0.616531	p3	0.328785
x4	0.871042	p4	0.079593

x1	0.120869	p1	0.116163
x2	0.354329	p2	0.443344
x3	0.621878	p3	0.354370
x4	0.872720	p4	0.086124

x1	0.124537	p1	0.098866
x2	0.368885	p2	0.401134
x3	0.631115	p3	0.401134
x4	0.875463	p4	0.098866

x1	0.084493	p1	0.076873
x2	0.256879	p2	0.333166
x3	0.473372	p3	0.364333
x4	0.714054	p4	0.185912
x5	0.907414	p5	0.039715

x1	0.085573	p1	0.071044
x2	0.261564	p2	0.317428
x3	0.476446	p3	0.377170
x4	0.715566	p4	0.193086
x5	0.907889	p5	0.041272

x1	0.087345	p1	0.061754
x2	0.268364	p2	0.283184
x3	0.484116	p3	0.401150
x4	0.719007	p4	0.209112
x5	0.908949	p5	0.044799

x1	0.089280	p1	0.051658
x2	0.275040	p2	0.239473
x3	0.500000	p3	0.417737
x4	0.724960	p4	0.239473
x5	0.910720	p5	0.051658

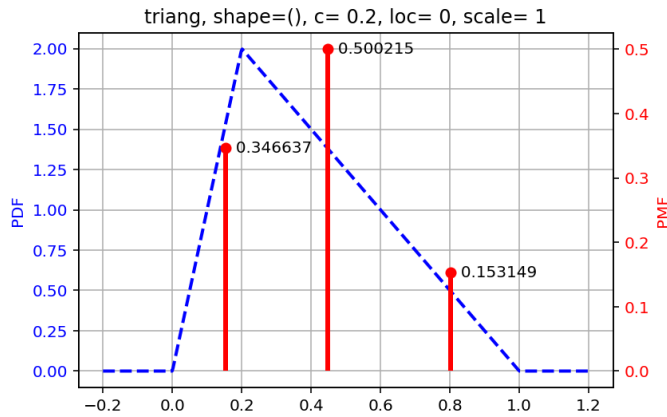
- For $0.5 < s \leq 1$, the approximate n -branch probability distribution is $\{(y_i, q_i)\}_{i=1}^n$

where $y_i = 1 - x_{n-i+1}$ $q_i = p_{n-i+1}$ for $i = 1$ to n

and $\{(x_i, p_i)\}_{i=1}^n$ is the approximate distribution for $\text{Triang}(0, 1, 1-s)$.

Example: $\text{Triang}(0, 1, 0.8)$ and $n = 3$.

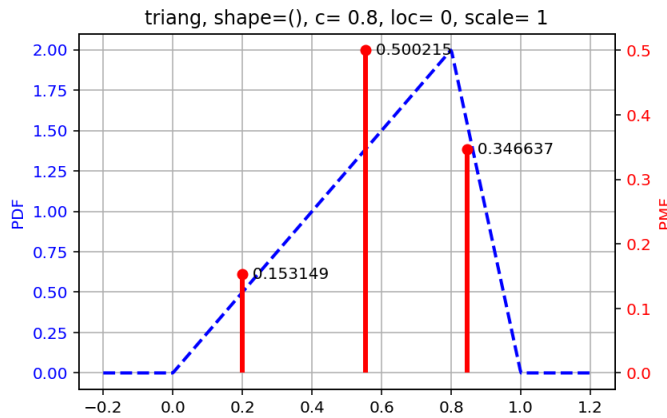
- To determine the 3-branch probability distribution for $\text{Triang}(0, 1, 0.8)$, we first get the results for $\text{Triang}(0, 1, 0.2)$, $n=3$:



x_1	0.154412	p_1	0.346637
x_2	0.447389	p_2	0.500215
x_3	0.801081	p_3	0.153149

- Next, transform the values to $\text{Triang}(0, 1, 0.8)$:

$$\begin{aligned} y_1 &= 1 - x_3 = 1 - 0.801081 = 0.198919 & q_1 &= p_3 = 0.153149 \\ y_2 &= 1 - x_2 = 1 - 0.447389 = 0.552611 & q_2 &= p_2 = 0.500215 \\ y_3 &= 1 - x_1 = 1 - 0.154412 = 0.845588 & q_3 &= p_1 = 0.346637 \end{aligned}$$



General Triangular (a, b, m) $a \leq m \leq b$

- location = a , scale = $(b - a)$, shape = $(m - a)/(b - a)$
- The n -branch discrete approximation for $\text{Triang}(a, b, m)$ is $\{(y_i, p_i)\}_{i=1}^n$

where $y_i = a + (b - a)x_i$ for $i = 1$ to n

and $\{(x_i, p_i)\}_{i=1}^n$ is the n -branch discrete approximation for $\text{Triang}(0, 1, (m - a)/(b - a))$

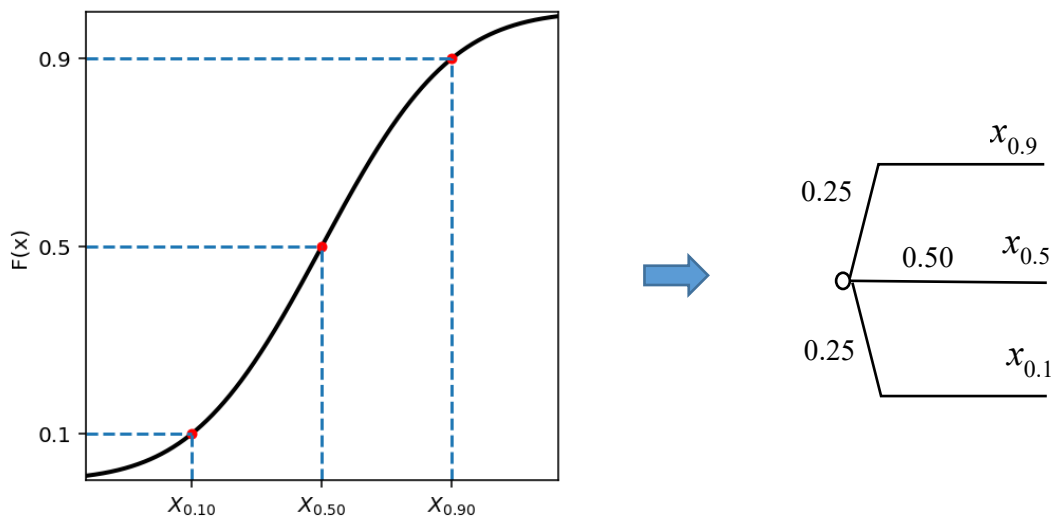
Discrete Approximation for Continuous Distributions in DPL

- In DPL, if a *discrete chance node* with n branches is created and a known standard continuous distribution is specified, DPL will automatically perform the $(2n - 1)$ moments matching and find the approximate branch values and probabilities for the decision tree. These values will be automatically used in creating the decision tree and generating the optimal policy tree.
- In DPL, if a *continuous chance node* is created and a known standard continuous distribution is specified, the resulting decision model can only be solved using Monte Carlo Simulation. There is no discrete optimal decision policy tree.

7.4.3 Three-Point Quick Approximation Methods

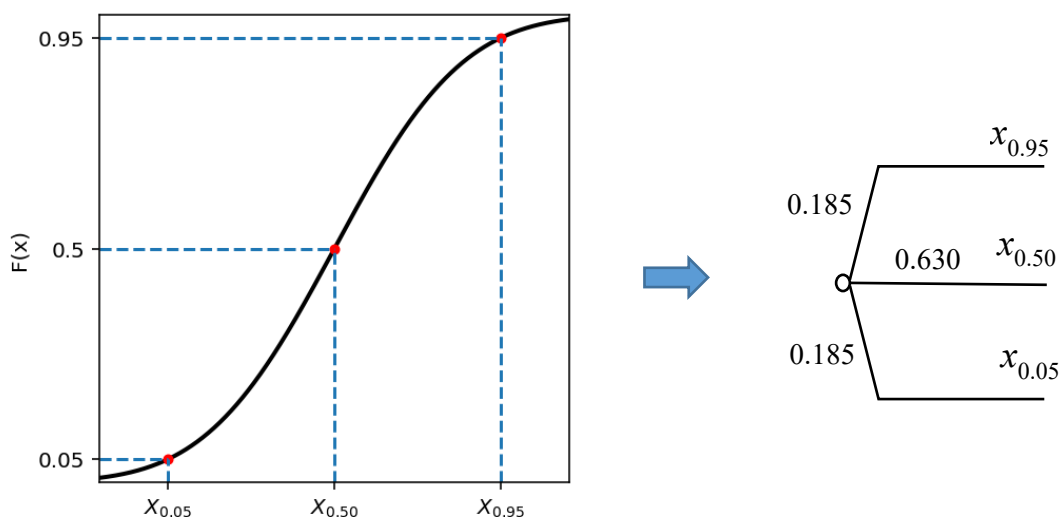
Stanford / SDG Approximation

- When the probability distribution is symmetrical, the 10, 50, and 90 percentiles provide a good approximation for a PMF with probabilities [0.25, 0.50, 0.25].



Pearson-Tukey Approximation

- When the probability distribution is symmetrical, the 5, 50, and 95 percentiles provide a good approximation for a PMF with probabilities [0.185, 0.630, 0.185].



References

1. A. Tversky and D. Kahneman (1974). Judgment under uncertainty: Heuristic and Biases. *Science* **185**:1124-1131. (Article 47 of Howard & Matheson, 1983).
2. A.C. Miller III and T.R. Rice (1983). Discrete Approximations of Probability Distributions, *Management Science* **29**(3):352-362.
3. D.L. Keefer and S.E. Bodily (1983). Three-Point Approximation for Continuous Random Variables, *Management Science* **29**(5):595-609.

Exercises

P7.1 Given the following data:

5.3299	4.2537	3.1502	3.7032	1.6070	6.3923
3.1181	6.5941	3.5281	4.7433	0.1077	1.5977
5.4920	1.7220	4.1547	2.2799		

- (a) Plot a histogram representing the data.
- (b) Fit a normal distribution to the data using any suitable method.
- (c) Plot the PDF of the fitted normal distribution and compare it with the plot in (a).

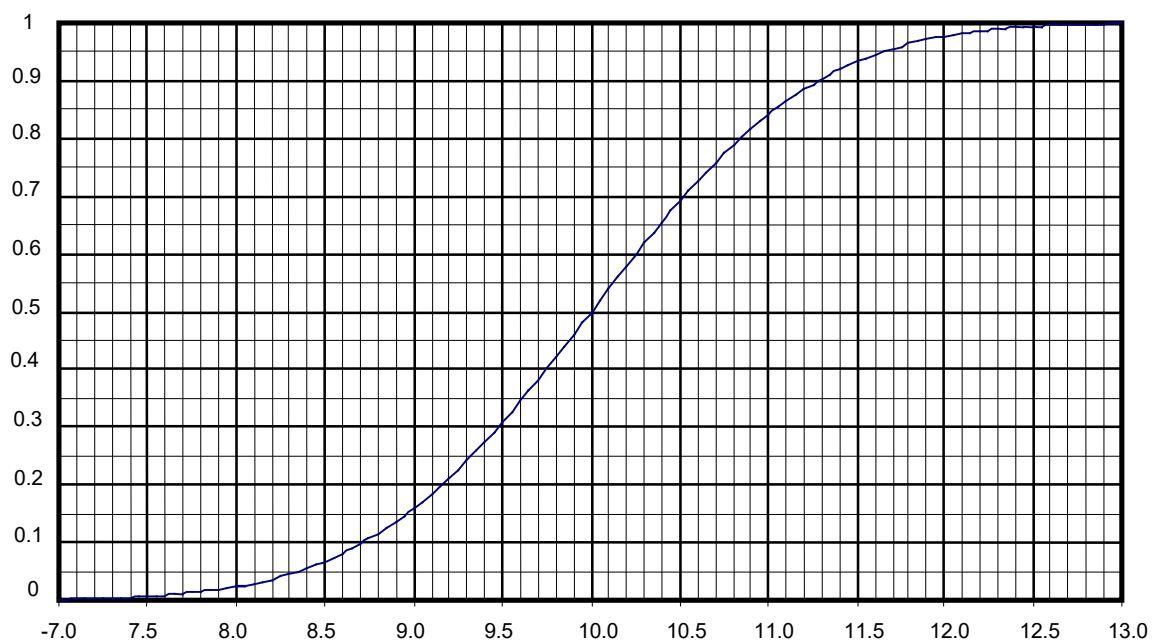
P7.2 Use DPL to find the discrete approximation of the following distributions with 3 branches:

- (a) A triangular distribution with $\min=0$, $\max=10$, $\text{mode}=5$.
- (b) An exponential distribution with $\text{mean}=10$.
- (c) A uniform distribution with $\min=10$ and $\max=20$.

P7.3 William faces an investment alternative whose net present value has a normal distribution with a mean of \$10 million and a standard deviation of \$1 million. William follows the Delta property and has a risk tolerance of \$5 million. Use DPL to determine and compare the certainty equivalent of this investment proposal using:

- (a) Discrete 3-branch approximation (moments matching).
- (b) Standard/SDG 3-branch quick approximation.
- (c) Pearson-Tukey approximation.

The CDF for the distribution Normal (10, 1) is given below:



P7.4 A scientist collected the following weights (in grams) of laboratory animals:

9.79	9.23	9.11	9.62
8.73	11.93	10.39	8.68
9.76	9.59	11.49	9.86
11.41	9.60	7.24	

- (a) Fit a probability distribution to the data using any suitable method.
- (b) Plot the PDF of the fitted distribution and compare it with the observed data.
- (c) Use the fitted distribution to estimate the probability that an animal's weight will be less than 9.5 grams.

P7.5 (Clement and Reilly 2001, Problem 10.11, modified)

A retail manager in a discount store wants to establish a policy regarding the number of cashiers to have on hand and also when to open a new cash register. The first step in this process is to determine the rate at which customers arrive at the cash register. One day, the manager observes the following times (in minutes) between arrivals of customers at the cash registers:

0.1	2.6	2.9	0.5
1.2	1.8	4.8	3.3
1.7	0.2	1.5	2.0
4.2	0.6	1.0	2.6
0.9	3.4	1.7	0.4

Assuming that customers' arrival is a Poisson process, fit an exponential distribution with one parameter λ to the data.

P7.6 (Clement and Reilly 2001, Problem 10.12, modified)

An ecologist studying the breeding habits of birds sent volunteers from the local chapter of the Audubon Society into the field to count nesting sites for a particular species. Each team was to survey five acres of land carefully. Because she was interested in studying the distribution of nesting sites within a particular kind of ecological system, the ecologists were careful to choose survey sites that were as similar as possible. In total, 24 teams surveyed five acres each and reported the following numbers of nesting sites in each of the five-acre parcels:

7	12	6	9
5	2	9	9
7	3	9	9
5	1	7	10
1	8	6	3
4	5	3	13

- (a) Plot a histogram for these data.
- (b) Fit a probability distribution to the data using any suitable method.
- (c) Compare the PMF of the fitted distribution with the histogram plotted in (a).